AP CALCULUS AB

Unit 1 • Lesson 1-1 • Guided notes

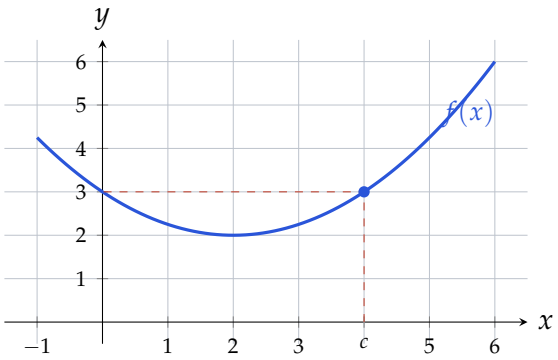
Lesson 1-1 • Limits to a Point on a Graph

Guided notes • Fill in the blanks and work the examples in the boxes

Objective: find left-hand, right-hand and two-sided limits from a graph.

Limit notation	From the left	From the right	From both sides
	_____ x approaches c with values _____ than c	_____ x approaches c with values _____ than c	_____ both one-sided limits are _____

My first limits



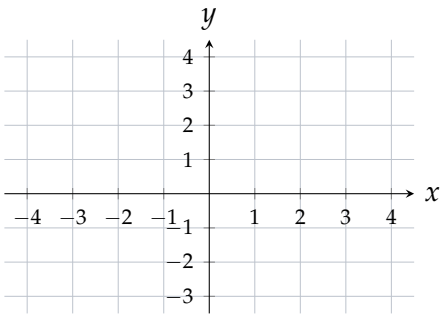
Use the graph, with $c = 4$.

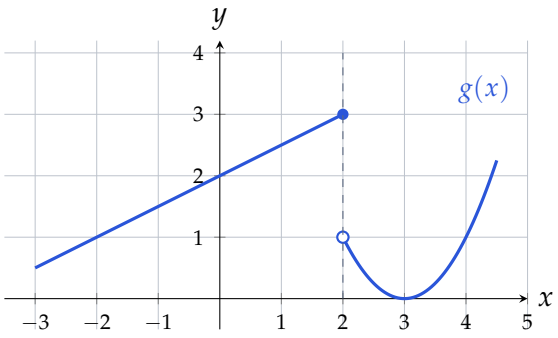
1) $\lim_{x \to 4^-} f(x) =$ ____ 2) $\lim_{x \to 4^+} f(x) =$ ____ 3) $\lim_{x \to 4} f(x) =$ ____

A limit is a _____ (a height): the height the graph _____ as x gets close to c.

Challenge

Draw the graph of a function whose left-hand and right-hand limits at $x = 1$ are **not** the same.



Example: one-sided limits are not the same	 1) $\lim_{x\to 2^-} g(x) =$ ____ 2) $\lim_{x\to 2^+} g(x) =$ ____ 3) $\lim_{x\to 2} g(x) =$ _____ 4) $g(2) =$ ____ (the height *at* $x = 2$: the filled dot) If the left and right limits are different, the two-sided limit _____ (DNE).
Summary	